

Class Design :

Question 1 :

Design a class called MarksAvg with the following data members and methods:

String name - To store the name of the student
int grade - To store the grade or class of the student
int english - To store the marks in english (out of 100)
int maths - To store the marks in math (out of 100)
int physics - To store the marks in physics (out of 100)
int chemistry - To store the marks in chemistry (out of 100)
double avg - To store the average of all the marks.

$$\text{Average of marks} = \frac{\text{Sum of marks in all subjects}}{\text{Total number of subjects}}$$

void input() - method to set the values for all variables
void calcAvg() - method to calculate the average marks of all subjects.
void display() - To print the name of the student, the class, and the average marks.

Write a main method to create an object and call the above methods

Question 2 :

Define a class called CarRent with the following data members and methods:

`int numDays` - To store the number of days the car is rented.

`int totalAmt` - To store the total cost for the car rented.

`void input()` - To input and store the necessary details in the variables.

`void amt()` - To calculate the total cost for the car rented

Use the following table to calculate the total cost depending on the number of days

Number of days	Cost per day (in ₹)
First 10 days	100
Next 20 days	80
Above 30 days	70

`void print()` - To print the total cost for the car.

Write a main method to create an object and call the above methods

Question 3 :

Define a class flightTicket to calculate the cost of a flight ticket. Use the following data members and methods.

String name - To store the name of the passenger

char ticketType - To store the type of ticket. e for economy, p for premium economy, f for first class.

int cost - To store the total cost of the ticket.

void userInput() - To take input from the user for the name and ticket type.

void calculate() - To calculate the cost of the ticket.

Economy - 2000 rupees

Premium Economy - 5000 rupees

First Class - 10,000 rupees

void display() - To display the cost of the ticket and the name of the passenger.

public static void main() - To create an object of the class and call the methods.